

Poojari Venkata Shreesh

AI/ML Engineer

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SUMMARY

AI/ML engineer passionate about building intelligent systems using knowledge graphs, natural language processing, and machine learning. I specialize in developing RAG-based applications, implementing logic programming solutions, and creating ML models that solve real-world problems. With experience in graph databases, LLM integration, and automated reasoning systems, I design solutions that combine cutting-edge AI technologies with practical applications. Currently pursuing a Master's in Computer Science at Arizona State University with focus on AI and Knowledge Representation, I'm eager to contribute to innovative AI projects that push the boundaries of what's possible.

EDUCATION

Arizona State University

Master of Science in Computer Science (MSCS) | GPA: 3.93

Tempe, AZ

Aug 2024 – Present

Relevant Coursework: Artificial Intelligence, Knowledge Representation and Reasoning, Machine Learning, Data Processing at Scale

BITS Pilani

Bachelor of Technology in Computer Science | GPA: 7/10

Hyderabad, India

Jul 2018 – May 2022

SKILLS

Languages: Python, C++, C, Java, JS, React JS

AI/ML Frameworks: Scikit-learn, NLTK, TensorFlow, PyTorch (familiarity)

Knowledge Graphs: Neo4j, RAG, LLM Integration, Google Gemini API, Ollama, Entity Extraction, Semantic Chunking

Logic Programming: Answer Set Programming (ASP), Clingo, Constraint Satisfaction

Databases: Neo4j, MongoDB, PostgreSQL

Tools: Python, Git, Linux, Shell (Bash/Zsh), Jupyter Notebooks, REST APIs

Other: Machine Learning, Artificial Intelligence, Natural Language Processing, Graph Algorithms, Multi-Agent Systems, Computer Vision, Data science, statistics

PROFESSIONAL EXPERIENCE

Darwinbox

Software Development Engineer

Hyderabad, India

Jul 2022 – Jul 2023

- Collaborated as part of the core/platform team and contributed towards two projects.
- Designed and developed features for workflow pages to include tagging and download options, improving usability.
- Developed dynamic data cards, implemented using MongoDB.
- Modularized legacy workflow modules to personalize data on a user level instead of a team level.

BlueJeans by Verizon

Frontend Intern

Bangalore, India

Jul 2021 – Dec 2021

- Collaborated on white box unit testing in the WMC team, resulting in a 5% increase in testing coverage.
- Developed unit test cases for the Web Meeting Client, increasing code testing coverage.
- Used Jest testing library to generate unit tests for ReactJS files and tracked progress using Git.

PROJECTS

AI Fashion Stylist App

Google Gemini API, Ollama, LLaVA, Computer Vision, Node.js, AI Chat

- Built an AI-powered fashion stylist MVP with Node.js/Express, integrating Google Gemini and local Ollama (Llama 3, LLaVA) for chat and image analysis with a unified service layer.
- Implemented multi-image clothing upload, AI-powered tagging (type, color, style), and occasion-based outfit recommendations using the user's wardrobe context.
- Designed REST API for wardrobe CRUD, AI chat, recommendations, and image analysis; added JWT auth, rate limiting, Helmet, and CORS for security and cross-origin support.
- Delivered a working web prototype with dark-mode UI, real-time AI chat with wardrobe context, and detailed error logging for failed image processing.

Accio: PDF/OCR Processing for Workflow Digitization

Neo4j, RAG, LLM, Knowledge Graphs, Python, NLP

- Developed a workflow digitization system that transforms PDF documents into structured Neo4j-backed knowledge graphs enriched with embeddings and exposed through a RAG interface.
- Integrated layout-aware OCR, semantic chunking, LLM triplet extraction, graph construction, and hybrid retrieval for grounded answer generation.
- Evaluated on 24 U.S. federal documents, populating a graph with 3,239 nodes and 4,941 relationships, achieving $F1 = 77.5\%$ for entity extraction and $NDCG@5 = 0.76-0.82$ for retrieval accuracy.

Automated Warehouse Scenario

Answer Set Programming, Clingo, Logic Programming, AI Planning

- Implemented an automated warehouse scenario using Answer Set Programming (ASP) where autonomous robots navigate a 2D grid, pick up shelves, and deliver products to picking stations.
- Modeled dynamic state transitions, enforced safety rules including collision avoidance, and developed plan optimization to minimize time steps using Clingo ASP solver.
- Extended logic to include shelf carrying actions, product deliveries, and concurrent multi-agent planning while ensuring spatial exclusivity and consistent state transitions.